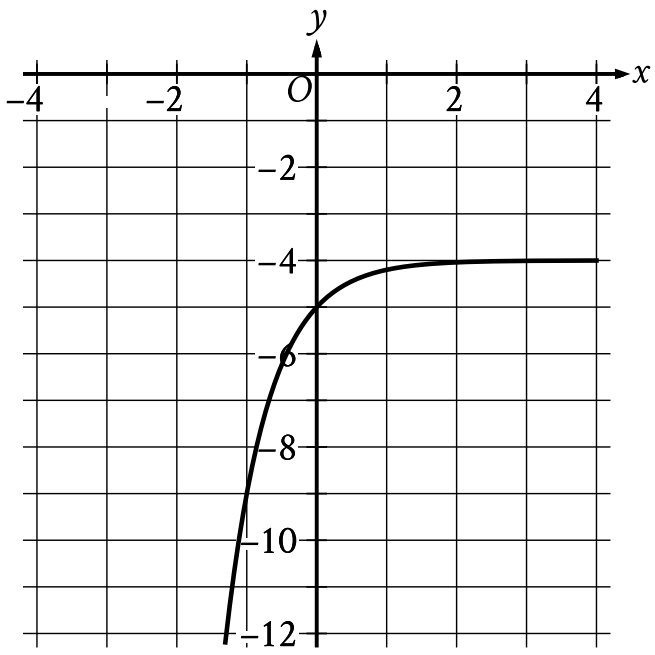
Question ID: 6abec9a8

Question



What is the y-intercept of the graph shown?

Answer

A. $(-1, -9)$

B. $(0, -5)$

C. $(0, -4)$

D. $(0, 0)$

Question ID: 788bfd56

Question

The function f is defined by $f(x) = 4 + \sqrt{x}$. What is the value of $f(144)$?

Answer

A. 0

B. 16

C. 40

D. 76

Question ID: b39d74a0

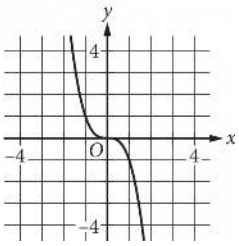
Question

x	y
0	0
1	1
2	8
3	27

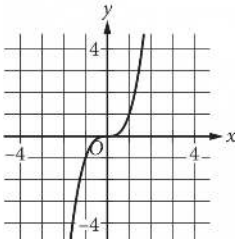
The table shown includes some values of x and their corresponding values of y. Which of the following graphs in the xy-plane could represent the relationship between x and y ?

Answer

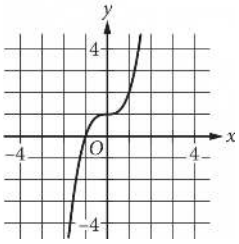
A.



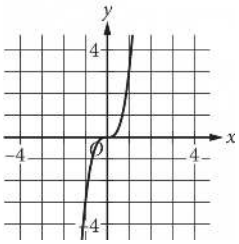
B.



C.

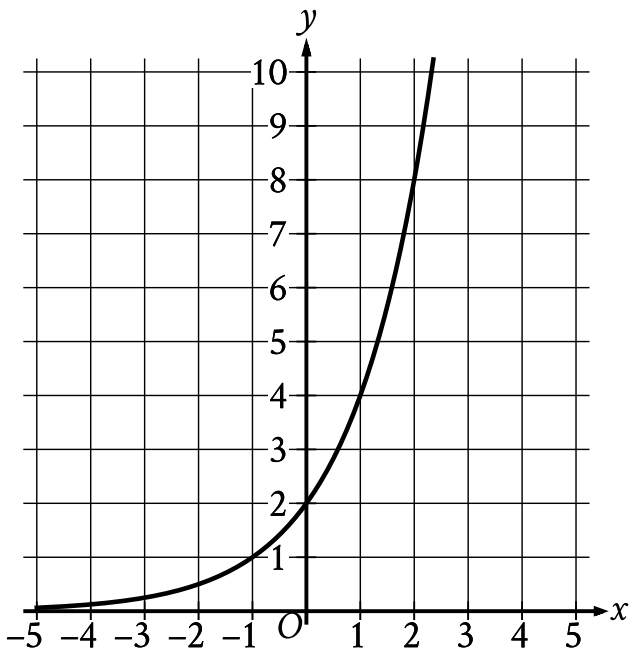


D.


y
4
x
−4
4
O
−4

Question ID: b5c43226

Question



What is the *y*-intercept of the graph shown?

Answer

A. $(0, 0)$

B. $(0, 2)$

C. $(2, 0)$

D. $(2, 2)$

Question ID: cfff8f8e

Question

At the time of posting a video, a social media channel had 53 subscribers. Each day for five days after the video was posted, the number of subscribers doubled from the number the previous day. Which equation gives the total number of subscribers, n, to the channel d days after the video was posted?

Answer

A. $n = (53)^d$

B. $n = 53(2)^d$

C. $n = 53\left(\frac{1}{2}\right)^d$

D. $n = (53)^2 + d$

Question ID: 5377d9cf

Question

If $f(x)=\frac{x^2-6x+3}{x-1}$, what is $f(-1)$?

Answer

A. −5

B. −2

C. 2

D. 5

Question ID: 75915e3c

Question

$f(x) = 2(3^x)$

For the function f defined above, what is the value of $f(2)$?

Answer

A. 9

B. 12

C. 18

D. 36

Question ID: 20722644

Question

The function f is defined by $f(x) = x^3 + 9$. What is the value of $f(2)$?

Answer

A. 14

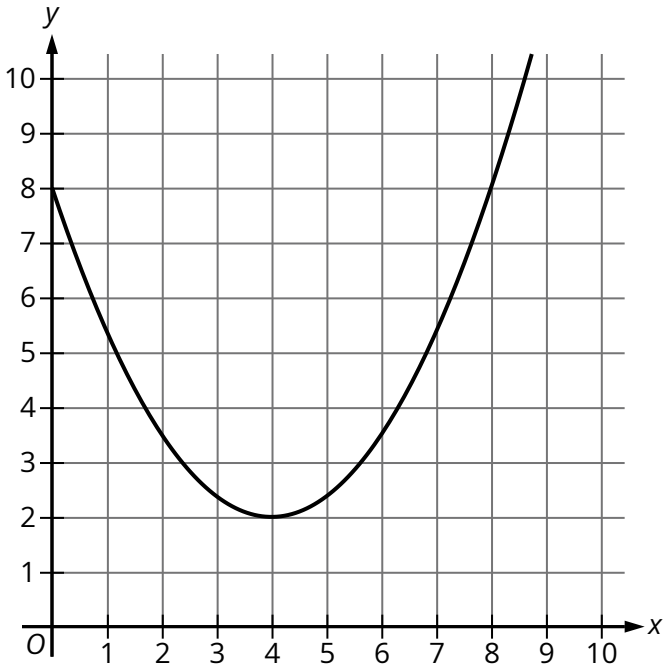
B. 15

C. 17

D. 18

Question ID: 5e63b9cf

Question



The graph shows a marble's height above the ground y, in inches, x seconds after it started moving on an elevated track of a marble run. Which of the following is the best interpretation of the y-intercept of the graph?

Answer

A. The marble's height was 0 inches above the ground 8 seconds after it started moving.

B. The marble's height was 8 inches above the ground when it started moving.

C. The marble's minimum height was 0 inches above the ground.

D. The marble's minimum height was 8 inches above the ground.

Question ID: d84a514a

Question

The function $f(x) = 240{,}000(1.22)^x$ gives a company's predicted annual revenue, in dollars, x years after the company started selling jewelry online, where $0 < x \leq 10$. What is the best interpretation of the statement "$f(5)$ is approximately equal to $648{,}650$" in this context?

Answer

A. 5 years after the company started selling jewelry online, its predicted annual revenue is approximately $648{,}650$ dollars.

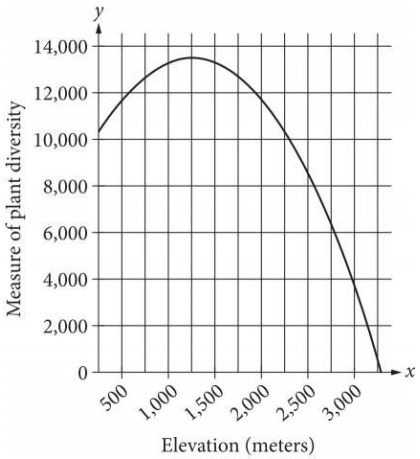
B. 5 years after the company started selling jewelry online, its predicted annual revenue will have increased by a total of approximately $648{,}650$ dollars.

C. When the company's predicted annual revenue is approximately $648{,}650$ dollars, it is 5 times the predicted annual revenue for the previous year.

D. When the company's predicted annual revenue is approximately $648{,}650$ dollars, it is 5% greater than the predicted annual revenue for the previous year.

Question ID: ebe4bde0

Question



The quadratic function graphed above models a particular measure of plant diversity as a function of the elevation in a region of Switzerland. According to the model, which of the following is closest to the elevation, in meters, at which plant diversity is greatest?

Answer

A. 13,500

B. 3,000

C. 1,250

D. 250

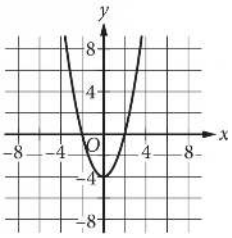
Question ID: d46da42c

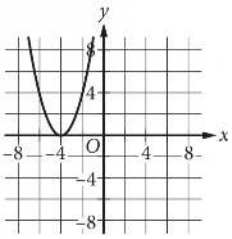
Question

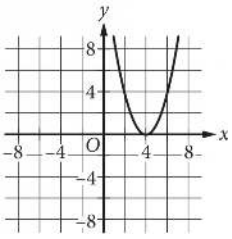
$f(x)=x^2+4$

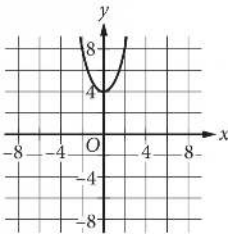
The function f is defined as shown. Which of the following graphs in the xy-plane could be the graph of $y=f(x)$?

Answer

A. 

B. 

C. 

D. 

Question ID: ee05c84e

Question

$f(x) = (x + 0.25x)(50 - x)$

The function f is defined above. What is the value of $f(20)$?

Answer

A. 250

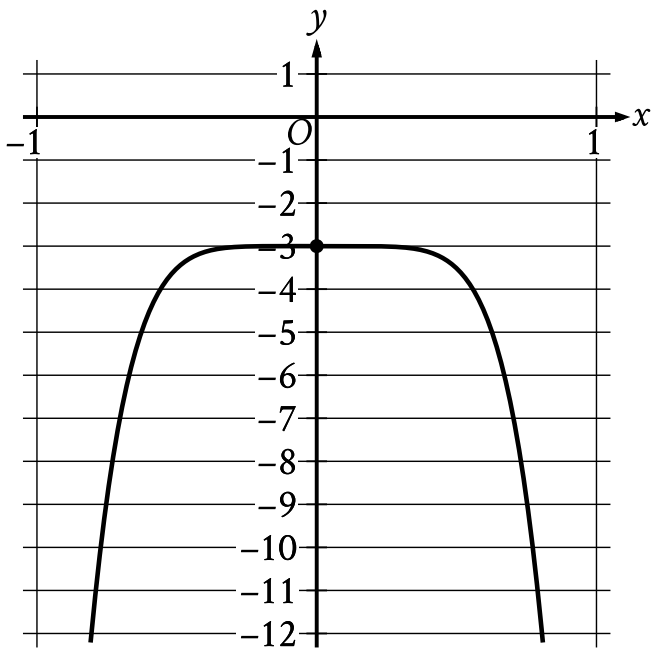
B. 500

C. 750

D. 2,000

Question ID: 50418728

Question



The graph of the polynomial function f, where $y = f(x)$, is shown. The *y*-intercept of the graph is $(0, y)$. What is the value of y?

Question ID: 79ba511a

Question

The function f is defined by $f(x) = x^3 + 15$. What is the value of $f(2)$?

Answer

A. 20

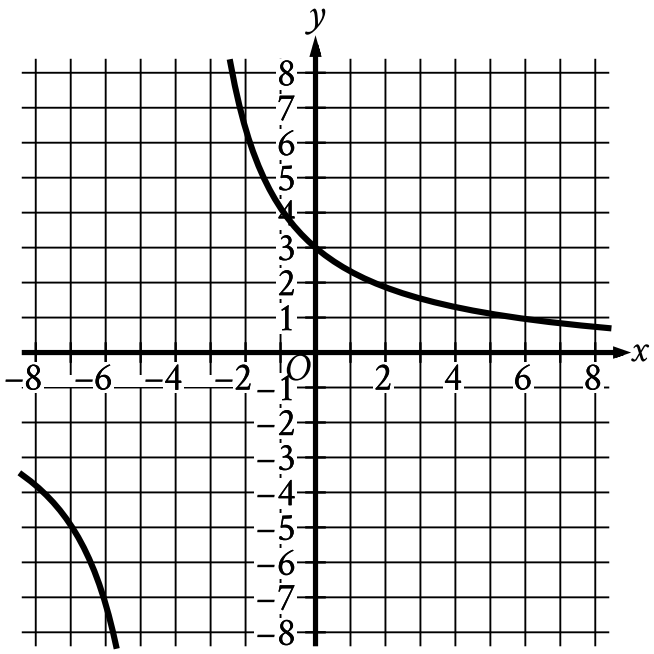
B. 21

C. 23

D. 24

Question ID: d45572cc

Question



The graph of $y = f(x)$ is shown in the *xy*-plane. The value of $f(0)$ is an integer. What is the value of $f(0)$?

Question ID: 0ad5012e

Question

$$y = -\frac{1}{4}x^2 + 2x + 29$$

The given equation models a company's scheduled deliveries over 8 months, where y is the estimated number of scheduled deliveries x months after the end of May 2012, where $0 \leq x \leq 8$. Which statement is the best interpretation of the *y*-intercept of the graph of this equation in the *xy*-plane?

Answer

A. At the end of May 2012, the estimated number of scheduled deliveries was 0.

B. At the end of May 2012, the estimated number of scheduled deliveries was 29.

C. At the end of June 2012, the estimated number of scheduled deliveries was 0.

D. At the end of June 2012, the estimated number of scheduled deliveries was 29.